

Lab 8: Fast Decay with RC Circuits

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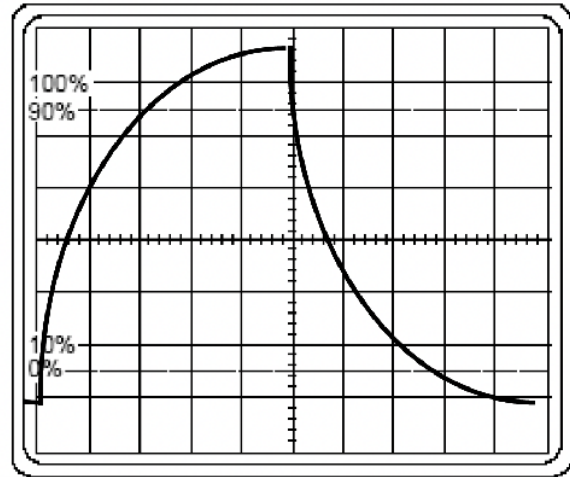
Professor Szwerc

Pre-lab Questions

1. You have constructed an RC circuit and you see the trace to the right on your oscilloscope screen. You have set the vertical scale to 1 V per square and the horizontal to 20 ms per square. What is the time constant of the RC circuit?

The time constant $\tau = RC$ is equivalent to the “1/e life” of the voltage across the capacitor. That is, at what time t after discharging does the voltage across the capacitor drop to $1/e \approx 36.8\%$ of its initial value?

The range of the voltage is from the line at -10% to the line at 110% . $1/e \approx 36.8\%$ between -10% and 110% is approximately 34.2% , which intersects with the curve about 1 division past the y-axis.



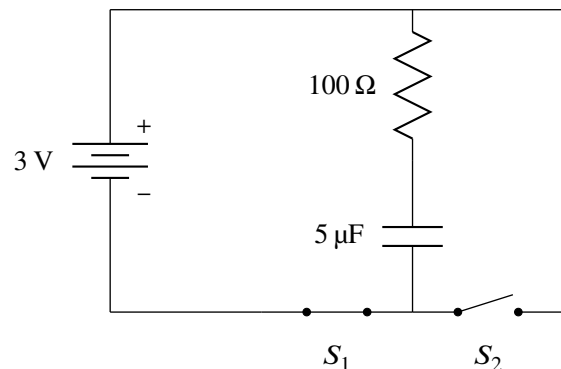
This corresponds to a time constant of approximately:

$$\tau \approx 1 \text{ division} \cdot 20 \text{ ms/division} = 20 \text{ ms.}$$

2. The RC circuit is constructed as shown on the right.
 - b. You then change the resistance to 200Ω . How does the time constant change?

The time constant is proportional to the resistance: $\tau \propto R$. Thus, doubling the resistance will double the time constant:

$$\tau' = 2\tau.$$



- b. You then change the capacitance to $20 \mu\text{F}$. How does the time constant change?

The time constant is proportional to the capacitance: $\tau \propto C$. Thus, quadrupling the capacitance will quadruple the time constant:

$$\tau' = 4\tau.$$

- c. You then change to a 9 V battery. How does the time constant change?

The time constant is independent of voltage, so the time constant will not change:

$$\tau' = \tau.$$

Procedure

1. An RC circuit was constructed with a $5600\ \Omega$ resistor and a $0.1\ \mu\text{F}$ capacitor.

Observed half-life:

$$t_{1/2, \text{observed}} = 1 \frac{\text{ms}}{\text{division}} \cdot 0.4 \text{ divisions} = 0.4 \text{ ms.}$$

Calculated half-life:

$$t_{1/2, \text{calculated}} = RC \ln 2 = 5600\ \Omega \cdot 0.1\ \mu\text{F} \cdot \ln 2 = 0.388 \text{ ms.}$$

Percent error: 3.05%

2. Switching resistance and capacitance:

- a. The old resistor is swapped with a $10\ 000\ \Omega$ resistor, and the same $0.1\ \mu\text{F}$ capacitor is used.

Observed half-life:

$$t_{1/2, \text{observed}} = 1 \frac{\text{ms}}{\text{division}} \cdot 0.6 \text{ divisions} = 0.6 \text{ ms.}$$

Calculated half-life:

$$t_{1/2, \text{calculated}} = RC \ln 2 = 10\ 000\ \Omega \cdot 0.1\ \mu\text{F} \cdot \ln 2 = 0.693 \text{ ms.}$$

Percent error: 13.44%

- b. The old capacitor is swapped with a $0.05\ \mu\text{F}$ capacitor¹, and the $10\ 000\ \Omega$ resistor from part (a) is used. We chose a capacitance that was half of the previous capacitance to “reverse” the effect of (approximately) doubling the resistance in part (a).

Observed half-life:

$$t_{1/2, \text{observed}} = 1 \frac{\text{ms}}{\text{division}} \cdot 0.36 \text{ divisions} = 0.36 \text{ ms.}$$

Calculated half-life:

$$t_{1/2, \text{calculated}} = RC \ln 2 = 10\ 000\ \Omega \cdot 0.05\ \mu\text{F} \cdot \ln 2 = 0.347 \text{ ms.}$$

Percent error: 3.87%

3. The resistor is replaced with a potentiometer with maximum resistance of $10\ 000\ \Omega$. At an arbitrary position of the potentiometer, the **observed half-life** is:

$$t_{1/2} \approx 0.5 \frac{\text{ms}}{\text{division}} \cdot 0.2 \text{ divisions} = 0.1 \text{ ms.}$$

From this observed half-life, we can **calculate the resistance** of the potentiometer at this position:

$$t_{1/2} = RC \ln 2 \rightarrow R = \frac{t_{1/2}}{C \ln 2} = \frac{0.1 \text{ ms}}{0.05\ \mu\text{F} \cdot \ln 2} = 2885\ \Omega.$$

Measured Resistance with DMM: $3360\ \Omega$ (15.20% difference)

¹Since there was no $0.05\ \mu\text{F}$ capacitor, we connected two $0.1\ \mu\text{F}$ capacitors in series to get an equivalent capacitance of $0.05\ \mu\text{F}$.

Conclusion

Examine the shapes of the exponential voltage signals across the resistor. Give a qualitative explanation for this shape in a manner similar to that presented above (in the METHOD section) for the RC circuit. Do these two voltages in fact add up to a square wave as demanded by Kirchhoff's rule? Explain your answer!

The voltage signals on the oscilloscope model exponential growth/decay, depending on if the square wave signal is high or low.

When the square wave is high, the capacitor is charged, so the voltage across the capacitor increases exponentially (more and more net charge builds up on the plates of the capacitor.) By Kirchhoff's loop rule, the voltage across the resistor + voltage across capacitor *must* sum to a constant voltage (until the square wave is low), so the voltage across the *resistor* will decay exponential such that the sum is constant.

An opposite effect is observed when the square wave is low: the capacitor discharges, so the voltage across the capacitor decreases exponentially, and thus the voltage across the resistor increases exponentially such that the sum is constant. Thus, the two voltages do add up to a square wave as demanded by Kirchhoff's rule.